

1. General Description

This project implements a media observer design pattern. Essentially, a dataset contains media items and notifies observers (players and viewers) affected by changes in these items. Media items are categorized into playable (audio, video) and non-playable (image, text).

2. Class and Interface Descriptions

Observer: This interface provides a structure for observers. It includes methods such as updating upon adding or removing media items, getting the observer's name, and displaying the list of observed media.

Dataset: This class represents a collection of media items and also saves observers. It provides methods for adding and removing media items, registering and removing observers, and obtaining lists of playable and non-playable media items. Additionally, it notifies registered observers of any changes in the dataset.

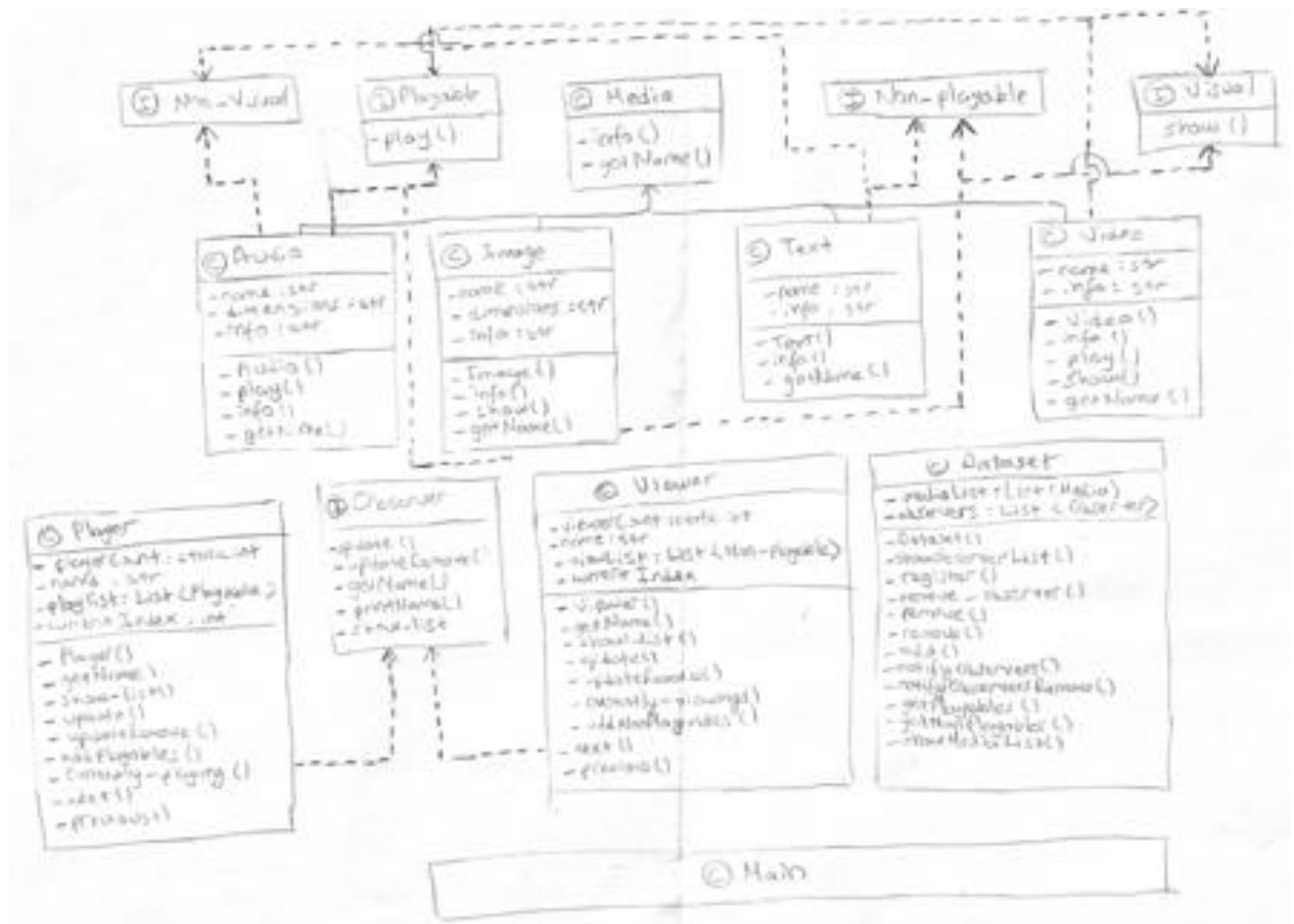
Player: This class implements the Observer interface and represents a player observing playable media items (e.g., audio and video). It manages a playlist of playable items and provides methods for adding, removing, and playing items. It considers only playable items while maintaining the playback sequence.

Viewer: This class implements the Observer interface and represents a viewer observing non-playable media items (e.g., images and text). It manages a list of viewed items and provides methods for adding, removing, and viewing items. It watches only viewable media items.

3. Flow of the Program

- The program begins with the main method.
- First, a dataset (Dataset) is created.
- Various media items (images, audio, video, text) are created and added to the dataset.
- Observers (players and viewers) are registered with the dataset.
- Observers, implementing the observer interface, wait to detect changes in media items.
- Addition or removal of media items is performed.
- Upon any change, the dataset notifies registered observers.
- Players manage playlists and play media items.
- Viewers manage lists and view non-playable media items.
- The program performs operations such as adding, removing, listing media items, and navigating between items (previous and next).
- At the end, the dataset removes registered observers and displays the updated list.
- The program completes its execution and exits.

UML Description



Output

```
busrasari@Busra-MacBook-Pro src % make run
java -cp ../classes Main
Observer registered: Player1
Observer registered: Player2
Observer registered: Viewer1
Observer registered: Viewer2

Media added to the dataset: imagename1
[Viewer1] New non-playable item added to the view list: imagename1
[Viewer2] New non-playable item added to the view list: imagename1

Media added to the dataset: imagename2
[Viewer1] New non-playable item added to the view list: imagename2
[Viewer2] New non-playable item added to the view list: imagename2

Media added to the dataset: imagename3
[Viewer1] New non-playable item added to the view list: imagename3
[Viewer2] New non-playable item added to the view list: imagename3

Media added to the dataset: imagename4
[Viewer1] New non-playable item added to the view list: imagename4
[Viewer2] New non-playable item added to the view list: imagename4

Media added to the dataset: imagename5
[Viewer1] New non-playable item added to the view list: imagename5
[Viewer2] New non-playable item added to the view list: imagename5

Media added to the dataset: audioname1
[Player1] New playable item added to the playlist: audioname1
[Player2] New playable item added to the playlist: audioname1

Media added to the dataset: audioname2
[Player1] New playable item added to the playlist: audioname2
[Player2] New playable item added to the playlist: audioname2

Media added to the dataset: audioname3
[Player1] New playable item added to the playlist: audioname3
[Player2] New playable item added to the playlist: audioname3

Media added to the dataset: videoname1
[Player1] New playable item added to the playlist: videoname1
[Player2] New playable item added to the playlist: videoname1

Media added to the dataset: videoname2
[Player1] New playable item added to the playlist: videoname2
[Player2] New playable item added to the playlist: videoname2

Media added to the dataset: videoname3
[Player1] New playable item added to the playlist: videoname3
[Player2] New playable item added to the playlist: videoname3

Media added to the dataset: textname1
[Viewer1] New non-playable item added to the view list: textname1
[Viewer2] New non-playable item added to the view list: textname1

Media added to the dataset: textname2
[Viewer1] New non-playable item added to the view list: textname2
[Viewer2] New non-playable item added to the view list: textname2

Media added to the dataset: textname3
[Viewer1] New non-playable item added to the view list: textname3
[Viewer2] New non-playable item added to the view list: textname3

List of Media in the Dataset:
Media List:
Image: imagename1, Dimensions: dimension info1, Info: other info1
Image: imagename2, Dimensions: dimension info2, Info: other info2
```

```
busra_sari_pa6

zsh - src + - - - x

List of Media in the Dataset:
Media List:
Image: imagename1, Dimensions: dimension info1, Info: other info1
Image: imagename2, Dimensions: dimension info2, Info: other info2
Image: imagename3, Dimensions: dimension info3, Info: other info3
Image: imagename4, Dimensions: dimension info4, Info: other info4
Image: imagename5, Dimensions: dimension info5, Info: other info5
Audio: audioname1, Duration: duration1, Info: other info1
Audio: audioname2, Duration: duration2, Info: other info2
Audio: audioname3, Duration: duration3, Info: other info3
Image: videoname1, Dimensions: duration1, Info: other info1
Image: videoname2, Dimensions: duration2, Info: other info2
Image: videoname3, Dimensions: duration3, Info: other info3
Text: textname1, Info: other info1
Text: textname2, Info: other info2
Text: textname3, Info: other info3

[Player1] Currently playing: audioname1
Audio: audioname1, Duration: duration1, Info: other info1

Media removed from the dataset: audioname1
[Player1] Playable item removed from the playlist: audioname1
[Player2] Playable item removed from the playlist: audioname1
[Viewer1] Non-playable item removed from the view list: audioname1
[Viewer2] Non-playable item removed from the view list: audioname1

[Viewer1] Currently viewing: imagename1
Image: imagename1, Dimensions: dimension info1, Info: other info1

[Viewer1] View List:
Image: imagename1, Dimensions: dimension info1, Info: other info1
Image: imagename2, Dimensions: dimension info2, Info: other info2
Image: imagename3, Dimensions: dimension info3, Info: other info3
Image: imagename4, Dimensions: dimension info4, Info: other info4
Image: imagename5, Dimensions: dimension info5, Info: other info5
Text: textname1, Info: other info1
Text: textname2, Info: other info2
Text: textname3, Info: other info3
Media removed from the dataset: imagename1
[Viewer1] Non-playable item removed from the view list: imagename1
[Viewer2] Non-playable item removed from the view list: imagename1

Updated List of Media in the Dataset:
Media List:
Image: imagename2, Dimensions: dimension info2, Info: other info2
Image: imagename3, Dimensions: dimension info3, Info: other info3
Image: imagename4, Dimensions: dimension info4, Info: other info4
Image: imagename5, Dimensions: dimension info5, Info: other info5
Audio: audioname2, Duration: duration2, Info: other info2
Audio: audioname3, Duration: duration3, Info: other info3
Image: videoname1, Dimensions: duration1, Info: other info1
Image: videoname2, Dimensions: duration2, Info: other info2
Image: videoname3, Dimensions: duration3, Info: other info3
Text: textname1, Info: other info1
Text: textname2, Info: other info2
Text: textname3, Info: other info3
Observer List:
Player1
Player2
Viewer1
Viewer2

[Player1] Playlist:
Audio: audioname2, Duration: duration2, Info: other info2
Audio: audioname3, Duration: duration3, Info: other info3
Image: videoname1, Dimensions: duration1, Info: other info1

[Player1] Playlist:
Audio: audioname2, Duration: duration2, Info: other info2
Audio: audioname3, Duration: duration3, Info: other info3
Image: videoname1, Dimensions: duration1, Info: other info1
Image: videoname2, Dimensions: duration2, Info: other info2
Image: videoname3, Dimensions: duration3, Info: other info3

Testing next and previous functionality:
Playing next audio:
Playing audio: audioname3
Playing previous video:
Error: No previous playable item of type video found.
Observer removed: Player1
Observer removed: Player2
Observer removed: Viewer1
Observer removed: Viewer2
Observer list is empty.
busrasari@Busra-MacBook-Pro src %
```